



Sustainability of biodegradable superabsorbent hydrogels in agriculture. A review

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Abstract

Biodegradable superabsorbent hydrogels have been proposed as soil conditioners to mitigate agricultural water stress, yet evidence on their agronomic relevance and environmental safety remains fragmented and contradictory. This review synthesizes current knowledge across five domains: material composition, swelling performance, biodegradation, field-scale application, and economic viability. A recurring performance–sustainability paradox is identified, whereby formulations that improve swelling stability and durability often compromise biodegradability, while environmentally benign designs show poor persistence and limited field efficacy. A critical analysis reveals a systematic overestimation of performance in laboratory studies relative to field conditions. Swelling capacities measured in distilled water are often 5–10 times higher than those realized in field soils, and biodegradation studies predominantly rely on indirect proxies rather than confirming complete mineralization, leaving ecological risks unresolved. Field application techniques, such as broadcasting, incorporation, and seed coating, provide localized benefits but lack scalability. Meanwhile, high production costs and incomplete life cycle assessments undermine claims of sustainability. Unlike earlier descriptive reviews, this study offers a critical integrative synthesis across technical, agronomic, ecological, and economic dimensions. Future progress depends on whole-system frameworks that incorporate standardized agro-climatic testing, isotopic tracing, and metagenomic monitoring of biodegradation, mechanized delivery technologies, and life cycle assessment. Such interdisciplinary validation is essential for harmonizing performance with sustainability and advancing BSAHs from experimental prototypes to credible tools for climate-resilient agriculture.

Keywords Material composition · Swelling performance · Biodegradation · Field-scale application · Economic viability

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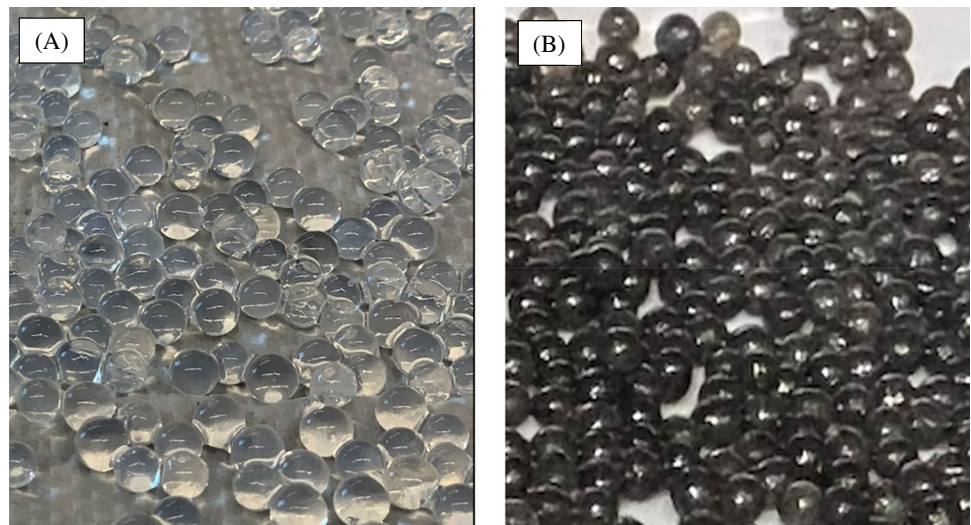
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Fig. 1 Biodegradable superabsorbent hydrogels without the addition of biochar (**A**) and with the addition of biochar (**B**). Photo credits: Sangchul Hwang (**A**), Portia Agyemang (**B**).



1 Introduction

Global agriculture is increasingly limited by climate variability, characterized by irregular rainfall patterns, intensifying droughts, rising surface temperatures, and accelerated evapotranspiration rates. These stresses lead to soil-water deficits, crop yield declines, and ongoing land degradation across diverse agro-ecological zones (Rita et al. 2020; Trnka et al. 2022; Brempong et al. 2023). Irrigation has become the primary adaptation strategy in arid and semi-arid regions; however, over-extraction of groundwater, salinization, and nutrient leaching threaten long-term soil productivity and ecosystem health (Fernández García et al. 2020; Scanlon et al. 2023). Therefore, new soil-conditioning technologies are urgently required to enhance water-use efficiency and maintain crop production amid climate uncertainty (Brempong et al. 2023).

Superabsorbent hydrogels have become a notable technology, capable of absorbing and retaining many times their weight in water, and slowly releasing it to crops during periods of moisture stress (Malik et al. 2022). However, conventional superabsorbent hydrogels are primarily produced from petroleum-based polymers, such as polyacrylamide and polyacrylate, which raise ecological concerns due to their persistent presence, microplastic formation, and potential toxicity (Li et al. 2025; V and Edison 2022). To address these issues, biodegradable superabsorbent hydrogels (BSAHs) have been developed (Fig. 1). In this review, BSAHs are defined by their functional capacity to undergo substantial microbial degradation in soil environments, ideally leading to mineralization. While commonly derived from renewable polymers such as starch, cellulose, alginate, chitosan, and agricultural residues (García-García; Nordin et al. 2024), this category also includes hybrid systems containing

synthetic components, provided the overall formulation is designed and demonstrated to be biodegradable. Das et al. (2025) reported that superabsorbent hydrogels provide a slow and sustained release of nutrients, helping to prolong their availability to plants. These materials also significantly improve the soil's water retention capacity. As they gradually break down, they leave behind non-toxic residues that integrate into the soil ecosystem, enhancing microbial activity and contributing to the buildup of soil organic matter (Kathi et al. 2021; Ni et al. 2023).

While prior reviews have examined the chemistry and laboratory-scale performance of BSAHs (Hasija et al. 2018; Solieman et al. 2023; Song et al. 2024), they have remained largely descriptive and have been restricted to pot-based or greenhouse experiments. These settings simplify soil structure, water dynamics, and microbial interactions, leading to an overestimation of BSAHs' real-world performance. What has been missing is a systematic and critical evaluation of BSAHs in *field-scale agricultural contexts*, where heterogeneous soils, fluctuating climatic conditions, and long cropping cycles may significantly influence their effectiveness. Without this perspective, current literature does not sufficiently inform their viability for adoption at the farm or landscape level.

This review, therefore, addresses an essential gap by critically examining the translation of BSAH technology from controlled environments to field-scale agriculture. It moves beyond simply reporting laboratory metrics to systematically evaluating performance under realistic field conditions, with an explicit focus on identifying and analyzing the disparities between controlled studies and real-world application. Specifically, this review (i) assesses the composition and modifications of BSAHs most suitable for large-scale deployment; (ii) analyzes their impacts on soil-water retention, plant

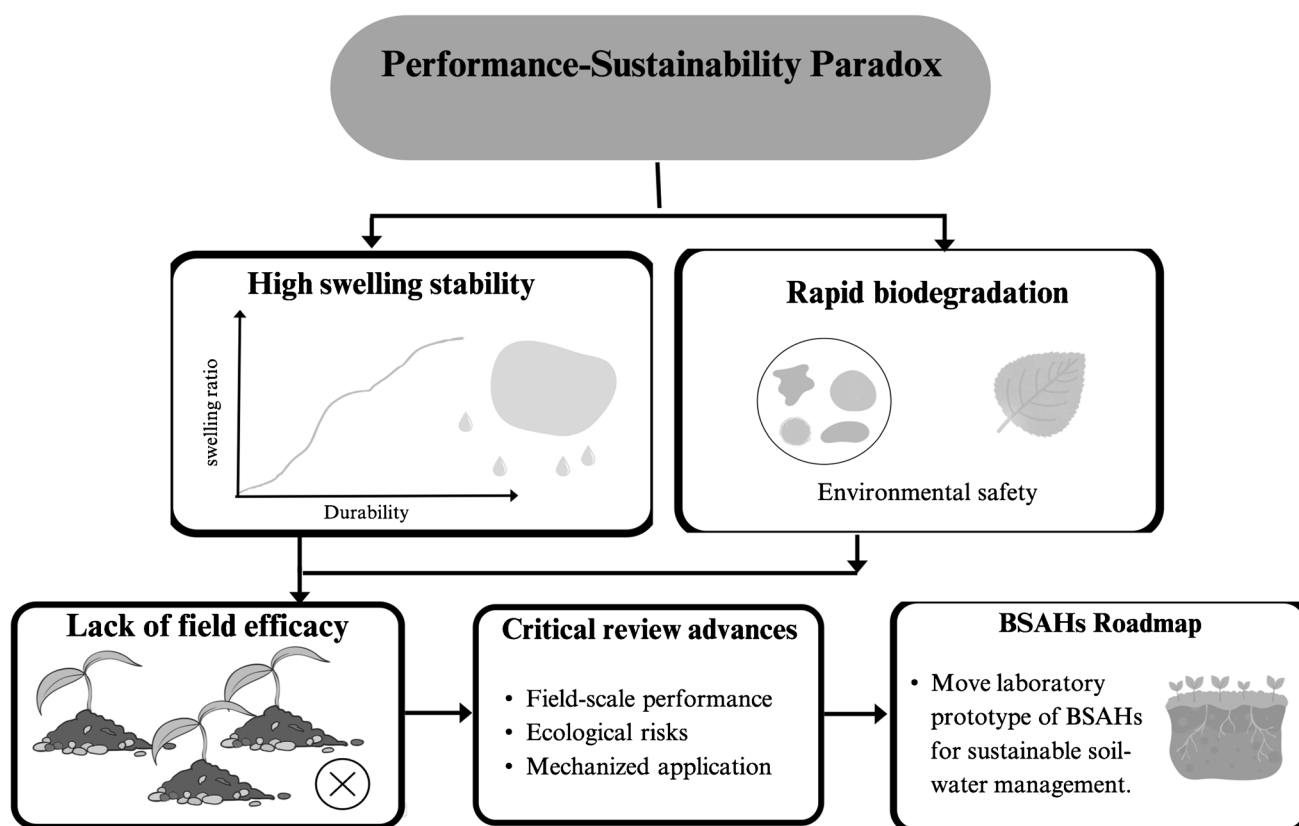


Fig. 2 Conceptual framework summarizing the performance–sustainability paradox in biodegradable superabsorbent hydrogels and the critical integrative advances highlighted in this review. Photo credit: Authors

growth, and agroecosystem health under variable field conditions, while critically juxtaposing these findings with laboratory results to highlight context-dependent performance gaps; (iii) critically examines biodegradation pathways, environmental risks, and life cycle implications; and (iv) identifies the technological, economic, and regulatory barriers to field implementation. In doing so, the review offers not only a synthesis of current knowledge but also an expert assessment of whether BSAHs represent a scientifically and ecologically viable strategy for climate-resilient agriculture. By explicitly situating this review within the existing body of literature and emphasizing where it advances a critical understanding of the lab-to-field translation, ecological risk assessment, and mechanized application, we aim to provide the agricultural and materials science community with a roadmap for progressing BSAHs from promising laboratory prototypes to robust tools for sustainable soil-water management. To visually situate this scope, Fig. 2 summarizes the performance–sustainability paradox of BSAHs, contrasting laboratory durability with biodegradability trade-offs, and highlighting how this review advances beyond descriptive summaries toward a critical synthesis of field performance, ecological risks, and mechanized applications.

2 Methodology and review strategy

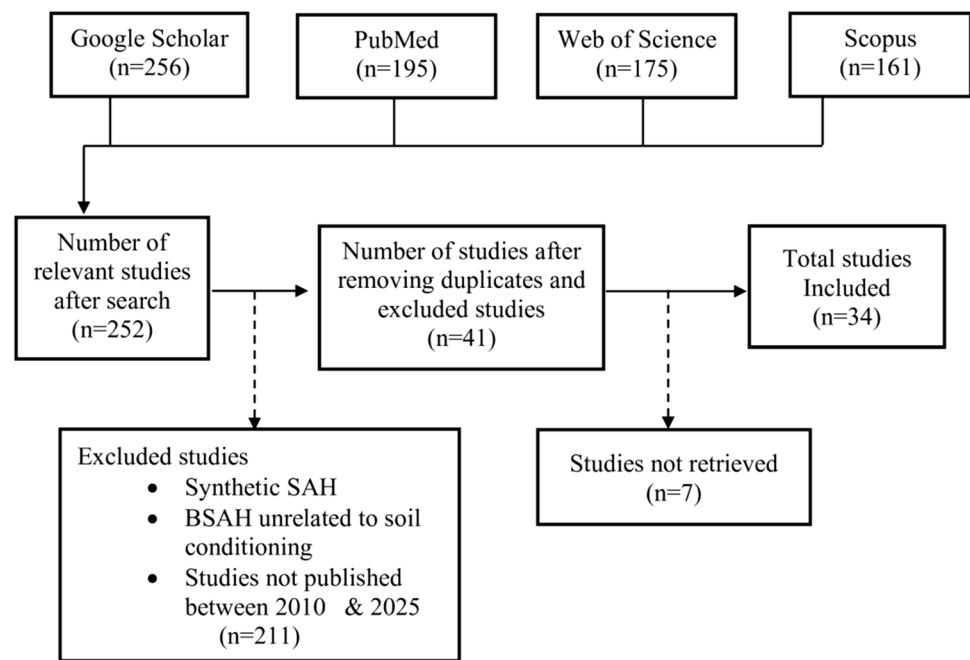
2.1 Rationale and design

Given the recent development of studies on superabsorbent hydrogels, a traditional narrative review would be inadequate to capture the complexity and diversity of BSAH research. To achieve both breadth and meaningful depth, this review employed a systematic scoping approach (Fig. 3) based on the framework by Munn et al. (2018), complemented by the PRISMA 2020 guidelines. Unlike earlier reviews that mainly summarize hydrogel synthesis methods or laboratory swelling performance (Hasija et al. 2018; Song et al. 2024), our methodology intentionally focused on field-oriented and scale-relevant studies, aiming to identify not only what has been tested but also where significant knowledge gaps persist regarding real-world agricultural applications.

2.2 Literature identification and search strategy

A comprehensive and systematic literature search was conducted across Scopus, Web of Science, PubMed, and Google

Fig. 3 PRISMA flow diagram of the systematic scoping review process. The diagram illustrates the identification, screening, and inclusion of studies focusing on biodegradable superabsorbent hydrogels for agricultural applications



Scholar from January 2010 to May 2025 to capture both foundational developments and recent innovations in BSAH research. The search strategy was designed to triangulate three core conceptual domains: (1) material identity (“biodegradable hydrogel”, “superabsorbent polymer”, “biodegradable superabsorbent hydrogel”, “BSAH”), (2) agricultural function (“soil water retention”, “soil conditioner”, “water-saving agriculture”, “drought mitigation”), and (3) application and fate (“field trial”, “biodegradation”, “environmental fate”, “soil amendment”).

These keyword strings were explicitly chosen to move beyond generic hydrogel reviews and specifically capture studies at the intersection of material science, agronomy, and environmental science. Boolean operators (AND, OR) were employed to construct targeted queries (e.g., (“biodegradable hydrogel” OR “BSAH”) AND (“soil” OR “agriculture”) AND (“water retention” OR “field trial”). The reference lists of key reviews and relevant primary articles were also hand-searched to ensure the inclusion of seminal and niche studies not captured by database algorithms.

2.3 Eligibility and selection of studies

The initial database search yielded over 800 records, which were subjected to a structured multi-stage screening process. The eligibility criteria were designed to capture the modern evolution of BSAHs, which is why the review focuses on literature published from 2010 onward. This temporal boundary is methodologically justified, as the research paradigm for agricultural hydrogels shifted decisively in the last decade toward explicitly addressing the performance–sustainability paradox (Li et al.

2025). Pre-2010 studies primarily investigated first-generation synthetic or minimally modified natural hydrogels, which do not reflect the advanced biopolymer chemistries, hybrid designs, and critical trade-offs between durability and biodegradability that define contemporary BSAH research and form the core of this review. To ensure foundational concepts from seminal earlier work were considered, the reference lists of all screened reviews and key articles were hand-searched, thereby incorporating influential pre-2010 literature that directly informs current material development. Based on this scope, the following inclusion and exclusion criteria were applied a priori:

2.3.1 Inclusion criteria

Studies were included if they were peer-reviewed primary research articles, technical reports, or doctoral theses published between 2010 and 2025, ensuring a focus on contemporary formulations and methods. Eligible materials were required to be hydrogels explicitly described as biodegradable or derived from renewable polymers (e.g., starch, cellulose) or hybrid systems with a dominant biodegradable component, aligning with the review’s objective to evaluate sustainable alternatives. The scope was restricted to agricultural or soil science applications reporting quantitative data on soil–water relations, crop response, hydrogel degradation in soil, or field-scale deployment.

2.3.2 Exclusion criteria

Studies were excluded if they focused solely on conventional, non-biodegradable synthetic hydrogels (e.g., pure

polyacrylamide) or on non-agricultural applications, such as biomedical or industrial uses. The exclusion of these studies was a deliberate methodological choice to maintain focus on the performance–sustainability paradox that is central to this review. While synthetic SAHs have a well-documented history of improving soil water retention, their persistence and potential for microplastic formation represent a fundamentally different environmental trajectory—one of managed persistence rather than designed biodegradation. Including them would conflate distinct material paradigms and obscure the critical trade-offs between durability and environmental fate that are unique to, and defining of, biodegradable formulations. Furthermore, purely synthetic or computational modeling studies, review articles without novel primary data, conference abstracts, and publications lacking sufficient methodological detail for critical appraisal were omitted to maintain the review's focus on empirical, applicative evidence.

After duplicate removal, titles and abstracts were screened for relevance, followed by full-text assessment against the predefined criteria. Low-quality records, including conference abstracts or reviews without primary data, were omitted. A total of 34 peer-reviewed documents were included in this study. The entire process was documented through a PRISMA flow diagram (Fig. 3) to ensure transparency, reproducibility, and methodological rigor.

2.4 Data extraction

For each eligible study, data were systematically extracted using a structured matrix designed to enable both comparison and critical synthesis across diverse contexts. Key variables captured included the type of biopolymer employed, the crosslinking strategy (chemical, enzymatic, or physical), and any hybrid additives such as biochar or nanoclays. Performance indicators were recorded with emphasis on swelling ratio, cyclic water retention, nutrient-holding capacity, and crop yield response. Soil–hydrogel interactions were examined through reported effects on texture compatibility, porosity, and microbial activity. Where available, degradation pathways were evaluated. In addition, details on application methods, such as broadcasting and seed coating, as well as assessments of environmental and economic implications, were catalogued. Critically, studies were categorized and tagged according to their experimental setting (laboratory, greenhouse/pot, small-plot, or field trial). This categorization was fundamental to the review's analytical approach, enabling a structured, comparative analysis to identify and interpret the recurring disparities between controlled-condition performance and field-scale outcomes. This framework provided a consistent basis for analyzing trends, identifying trade-offs, and evaluating the scalability and ecological safety of BSAHs.

2.5 Critical synthesis approach

To move beyond a descriptive summary of existing studies, the evidence was synthesized through a comparative and critical lens. Findings were systematically contrasted to highlight divergences in hydrogel performance across soil types, climatic conditions, and most importantly, experimental scales. Particular attention was given to identifying paradoxes and trade-offs, such as the enhanced water-retention capacity achieved through synthetic grafting versus the ecological risks associated with reduced biodegradability. A core component of the synthesis involved directly comparing laboratory-derived claims (e.g., swelling capacity, degradation rate) with outcomes from field trials to quantify and explain the frequently observed performance gaps. Greater interpretive weight was assigned to field trials and long-term studies, while laboratory and pot-based results were critically examined for their limitations in extrapolation to real agricultural conditions. Throughout the synthesis, emphasis was placed on integrating agronomic outcomes with environmental implications, thereby providing not only a consolidated overview of performance metrics but also an expert assessment of their scalability, ecological safety, and long-term viability in agricultural systems.

3 Results and discussion

3.1 Composition and modification of BSAHs for soil conditioning

BSAHs have been primarily developed from renewable biobased biopolymers such as corn starch, cellulose, chitosan, alginate, and lignin (Fig. 4), often blended with agricultural residues like rice husk or corn stover (García-García et al. 2024; Hasija et al. 2018). Among these, alginate, a natural polysaccharide derived from brown seaweed, is particularly prominent due to its inherent hydrophilicity, biocompatibility, and ability to form gentle ionic gels with divalent cations such as Ca^{2+} , making it a preferred backbone for numerous moisture-retention and seed-coating applications (Skrzypczak et al. 2021; Ghobashy 2020). While lignin, a complex aromatic polymer from plant cell walls, offers potential as a low-cost, rigidifying agent, its use in BSAHs is less advanced, primarily due to challenges with solubility and consistent chemical functionality. These polymers offer inherent biodegradability and compatibility with soil ecosystems; however, their performance under field conditions is often limited by their restricted swelling capacity, poor structural stability, and susceptibility to rapid microbial degradation. For example, pure alginate hydrogels, while environmentally benign, often suffer from low mechanical strength and rapid dissolution in aqueous environments, necessitating modification for durable field

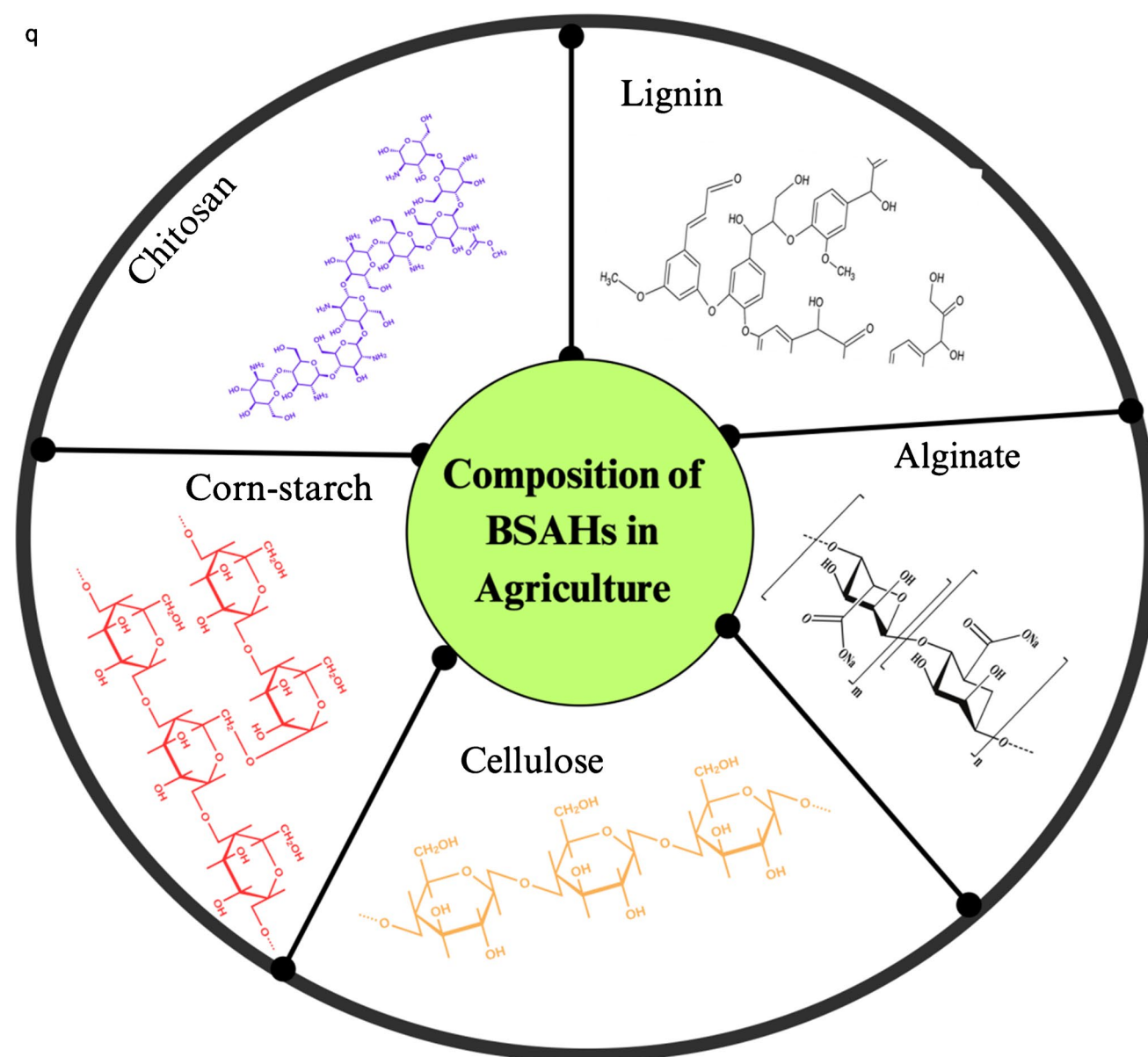


Fig. 4 Chemical structure of some renewable biopolymers for biodegradable superabsorbent hydrogels development. Photo credit: Authors

use. To overcome these weaknesses, numerous studies have introduced synthetic grafts like polyvinyl alcohol, acrylic acid, and acrylamide (Bora et al. 2023; Pettinelli et al. 2024; Solieman et al. 2023; Sharma et al. 2015) or chemical crosslinkers such as N, N'-methylenebisacrylamide and ethylene glycol dimethacrylate to reinforce the hydrogel matrix and enhance water-holding performance (Jungsinyatam et al. 2022; Kenawy et al. 2021; Rop et al. 2019; Saleem et al. 2024; Thombare et al. 2018).

However, these strategies expose a fundamental paradox. The very modifications that improve swelling and durability compromise the environmental rationale for BSAHs, raising concerns over incomplete biodegradation

and potential microplastic formation (Pettinelli et al. 2024; Thombare et al. 2018). While proponents argue that even partial substitution of synthetic components with natural polymers reduces ecological risks, evidence from soil incubation trials suggests that residual acrylamide-based fragments may persist for far longer than cropping cycles (Kenawy et al. 2021). Such findings underscore the urgent need for rigorous, long-term degradation studies rather than reliance on short-term weight loss or CO₂ evolution as proxies for complete mineralization.

Recent advances have sought to address this challenge by investigating biobased crosslinkers such as citric acid, tannins, or genipin, as well as combining them with biochar or

nanoclays to enhance water retention and biodegradability at the same time (Das and Ghosh 2022; García-Gómez et al. 2024; Iqbal et al. 2024; Turioni et al. 2021). Although promising, these methods are still mostly tested in laboratories or pot trials, and their scalability in diverse agricultural soils remains unverified. Taken together, the current state of BSAH design reveals a critical tension between functional performance and ecological integrity. The field must move beyond incremental chemical grafting strategies and embrace integrative approaches that prioritize green crosslinking, renewable additives, and field-validated testing protocols. Without such a shift, BSAHs may remain promising laboratory prototypes that fail to deliver sustainable solutions at the scale of agricultural practice.

3.2 Physicochemical properties and swelling behavior in agricultural soils

One of the most widely reported attributes of BSAHs is their exceptionally high swelling capacity, with values exceeding 604% (Guo et al. 2022) and 420% in distilled water conditions (Iqbal et al. 2024). However, these figures are frequently misinterpreted as direct predictors of field performance, a common oversimplification in the literature that neglects critical soil-mediated constraints (Guilherme et al. 2015; Saha et al. 2020; Malik et al. 2022; Agbna and Zaidi 2025). Ionic strength, soil compaction, bulk density, and capillary pressures can reduce swelling capacity by an order of magnitude, often bringing it below 94.88% (Teng et al. 2024) and 26% (Palma et al. 2024) in real soil environments. This discrepancy raises critical questions about the reliability of laboratory metrics as predictors of agronomic impact.

Additionally, several studies (García-Gómez et al. 2024; Kenawy et al. 2021; Pathak and Ambrose 2020; Teng et al. 2024) have demonstrated that while BSAHs improve short-term soil moisture retention and seedling emergence, their performance varies drastically across different soil textures and salinity gradients. For example, BSAHs exhibit appreciable water-holding improvements in loamy soils (Amanya et al. 2025; Seybold 1994). However, their agronomic efficiency, defined here as the translation of water retention into reliable plant-available moisture and yield benefit, can be inconsistent in sandy soils. While some studies report high swelling magnitudes in sandy textures due to low mechanical confinement, the resultant moisture can be susceptible to rapid drainage or insufficient root-hydration contact, limiting practical benefit (Palma et al. 2024). In saline soils, osmotic potential directly suppresses hydrogel water uptake, reducing retention capacity (Guo et al. 2022). Such context-dependence underscores that general claims of “improved water retention” are scientifically misleading without specifying the soil matrix in which hydrogels are applied, as reported performance gains are frequently specific to particular soil textures and do not extrapolate

universally (e.g., Kathi et al. 2021; Palma et al. 2024). Beyond this spatial variation, a critical temporal limitation arises from repeated wetting–drying cycles, which induce polymer fatigue and progressively reduce swelling capacity—a key durability concern (Teng et al. 2024). However, measurements of this cyclic behavior, like those of initial swelling, are predominantly conducted under controlled conditions (e.g., pot, column studies) across limited soil types. Consequently, the long-term performance fatigue of BSAHs in diverse, real-world field soils remains poorly quantified, severely limiting the extrapolation of these mechanistic insights to field-scale predictions.

Although controlled studies produce encouraging signals, their evidentiary value for field-scale application is limited. Experimental systems such as the one shown in Fig. 5—where tomato plants are grown in BSAH-amended soil in plastic cups—fail to replicate field conditions. Such setups cannot reproduce critical field variables, including soil bulk density gradients, macropore networks, root-soil aggregate structure, or the overburden stresses that constrain hydrogel expansion and root-water dynamics in situ. Laboratory swelling in unconstrained or lightly constrained conditions could overestimate accessible water in intact field soils, where compaction and overburden could reduce effective BSAH expansion and thus available capillary or absorbed water. The extrapolation from cup or small-plot experiments that report tens to hundreds g/g swelling to volumetric water content improvement at the field scale is therefore mechanistically weak, a common but often unvalidated practice in the literature that overstates potential agronomic benefits (e.g., Guilherme et al. 2015). Notably, field studies on related biopolymer soil amendments have demonstrated sustained improvements in vegetation growth and soil strength over multiple seasons (Ni et al. 2023), suggesting that durable benefits are possible but require formulations and testing under realistic field conditions. The evidence suggests that the physicochemical performance of BSAHs in soils is less a question of “maximum swelling capacity” and more about the interaction between polymer structure, soil matrix, and environmental stressors. Some of these discrepancies are summarized in Table 1. A critical research priority should therefore be the development of standardized testing protocols that evaluate swelling behavior under realistic soil–climate conditions across scale, culminating in field validation, rather than idealized laboratory systems. Without this shift, the literature risks perpetuating a cycle of overestimated laboratory claims that fail to translate to the scale of practical agriculture.

Discrepancies between laboratory-reported swelling capacities and performance in soil-based studies (pot, column, greenhouse) of biodegradable superabsorbent hydrogels (BSAHs) are presented in Table 1. The table highlights the performance drop in soil environments but underscores

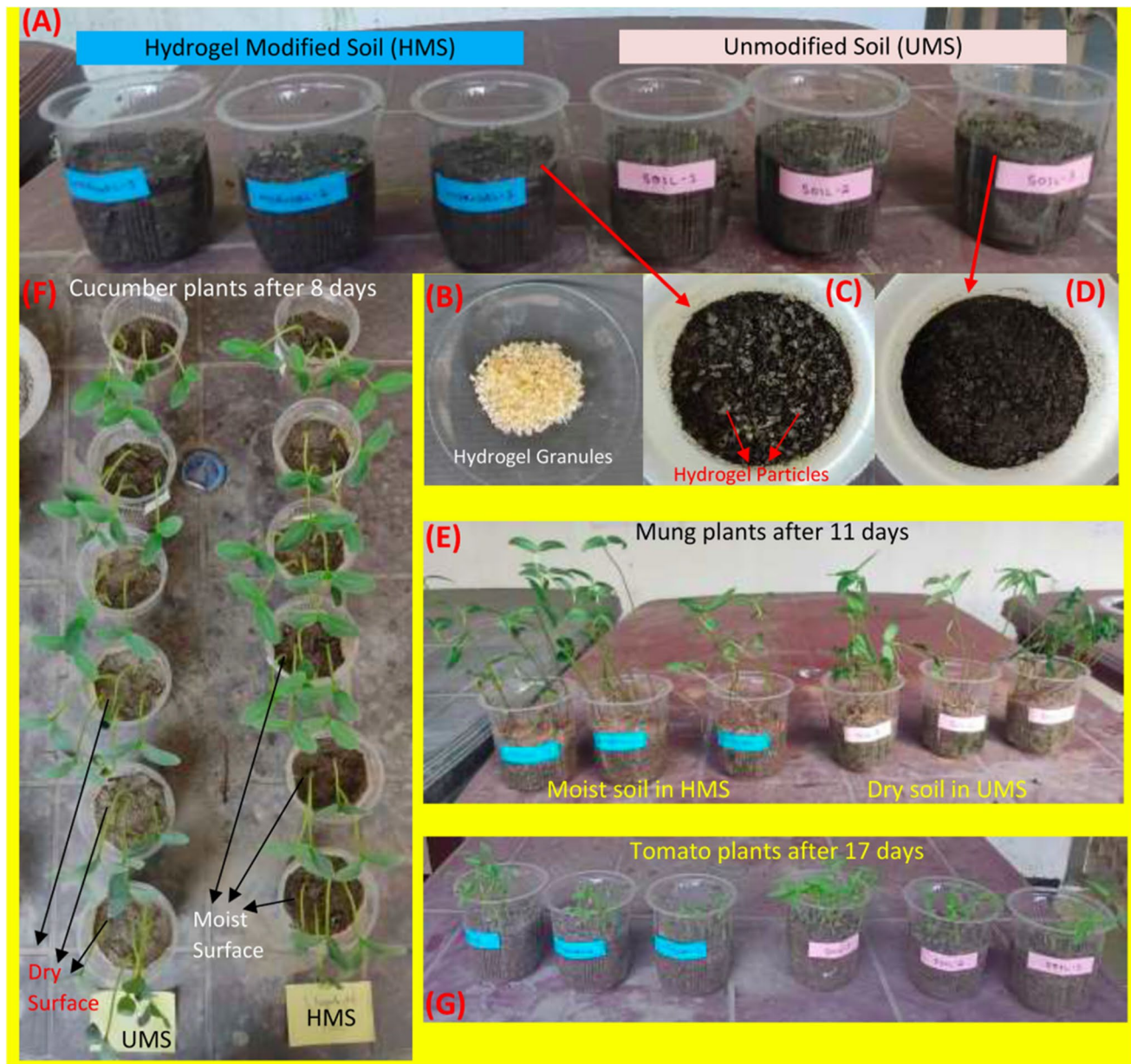


Fig. 5 Soil conditioned with biodegradable superabsorbent hydrogels to improve tomato growth in experimental plastic cups: **A** Pots of HMS and UMS, **B** hydrogel granules (~2 mm), **C** top view of HMS, **D** top view of UMS, **E** growth of mung plants in HMS and UMS

after 11 days, **F** growth of cucumber plants in HMS and UMS after 8 days, and **G** growth of tomato plants in HMS and UMS after 17 days. (Adapted with permission from Elsevier, Das et al. 2025).

that most data are not from agronomic field trials, limiting direct field-scale inference.

3.3 Conceptual and mechanistic challenges limiting field performance

The well-documented attenuation of BSAH swelling capacity in soil versus distilled water is a diagnostic symptom of a deeper, multi-factorial challenge (Saha et al. 2020). Moving beyond this empirical observation requires identifying

the interconnected conceptual and mechanistic barriers that constrain field efficacy. These barriers arise from fundamental mismatches between the idealized conditions of material synthesis and the complex, heterogeneous, and dynamic soil environment. Four primary domains of challenge are critical:

3.3.1 Physical confinement and soil structure

The swelling of a hydrogel is a volumetric expansion that must displace the surrounding soil matrix. Under field

conditions, soil overburden pressure, compaction, and the presence of rigid aggregates create a confining stress that physically restricts expansion (Palma et al. 2024). Furthermore, the hydrogel's location within the pore network is critical; granules deposited in macropores may swell without significant confinement but have poor contact with the soil-water menisci and plant roots, limiting hydraulic connectivity and water release to plants.

3.3.2 Chemical and osmotic limitations

Soil solution is not pure water. The presence of multivalent cations (e.g., Ca^{2+} , Mg^{2+}) can induce ionic crosslinking within anionic hydrogel networks, prematurely “locking” the polymer and reducing its water uptake capacity (Guo et al. 2022). In saline soils, high ionic strength creates an unfavorable osmotic gradient, severely limiting the hydrogel's ability to absorb water against the external salt concentration.

3.3.3 Dynamic Hydrological and Biotic Interactions

Field soils undergo repeated wetting and drying cycles. Each cycle can cause mechanical fatigue in the hydrogel network, leading to progressive breakdown of crosslinks and irreversible loss of re-swelling capacity (cyclic fatigue) (Teng et al. 2024). Concurrently, the intended biodegradation process begins immediately upon soil incorporation. The timing of this microbial digestion is often misaligned with crop water demand; significant structural degradation can occur during the critical growth period when water retention is most needed (Kenawy et al. 2021).

3.3.4 Spatial heterogeneity and application inefficiency

Standard application methods (e.g., broadcasting) result in a sparse, non-uniform distribution of hydrogel granules in the root zone. This spatial discontinuity creates isolated pockets of moisture without forming a continuous, plant-accessible water reservoir. The probability of a root encountering a swollen hydrogel granule is often low, decoupling the material's water-holding capacity from actual plant water uptake, as observed in field trials (AbdAllah et al. 2021).

These interconnected challenges are synthesized in Fig. 6, which provides a conceptual model of how soil-physical, chemical, biological, and methodological factors collectively constrain the translation of intrinsic BSAH water absorption capacity into realized agronomic water-use efficiency. Addressing this performance gap requires a systems-level design approach that engineers BSAHs not for maximum swelling in water, but for optimal function within the specific physical-chemical-biological continuum of the target soil ecosystem.

3.4 Biodegradation pathways and environmental fate

The environmental promise of BSAHs rests on their presumed complete biodegradation in soil, yet the mechanisms and endpoints remain inadequately characterized. Primary degradation mechanisms include hydrolysis of ester/glycosidic bonds, enzymatic cleavage by soil microbes (e.g., cellulases, chitinases), and oxidation. However, the loss of effectiveness—the decline in water-holding capacity—often precedes and proceeds independently of mass loss, driven by polymer network fatigue, ionic crosslinking from soil cations, and microbial surface colonization that blocks water ingress (Kenawy et al. 2021; Teng et al. 2024). Laboratory soil incubation studies frequently report partial mass loss (e.g., ~20% in 120 days for alginate-based hydrogels), but rely on indirect proxies (weight loss, CO_2 evolution) that cannot distinguish between complete mineralization and fragmentation into persistent oligomers or microplastic residues—a critical ecological distinction (Zhou et al. 2025).

Reported degradation rates vary dramatically with material composition and environmental conditions, often obscuring field-relevant predictability. For example, Pettinelli et al. (2024) observed 85% mass loss of a sodium alginate-g-polyacrylamide hydrogel in soil within 5 weeks. This relatively rapid breakdown may align with short-season crops but risks premature loss of water-retention function. In contrast, Saleem et al. (2024) reported 95.65% biodegradation of a Guggul-gum-based hydrogel within only 3 h in vermicompost, illustrating extraordinarily rapid degradation in a biologically enriched, non-field medium. Such extreme variability underscores that high biodegradability in optimized laboratory or composting systems does not guarantee predictable, agronomically useful degradation kinetics in field soils, where temperature, moisture, and microbial activity fluctuate widely. These examples reinforce the necessity of evaluating degradation not merely by mass loss but by the alignment of functional longevity with crop water demand and the absence of persistent, potentially harmful residues.

Field evidence is even more limited; recent trials with starch-based hydrogels used as seed coatings demonstrated measurable breakdown over cropping cycles, but the fate of intermediate degradation products in soil microbial communities remains unresolved (Rakhmetullayeva et al. 2025). This knowledge gap is critical, as oligomer accumulation could alter soil pore structure, influence nutrient cycling, or disrupt rhizosphere microbial balance. Without isotopic tracing or metagenomic profiling, claims of environmental safety risk are premature.

Another consideration is the altered degradation profile of chemically modified, hybrid BSAHs. As noted in Sect. 3.1, incorporating synthetic grafts, such as acrylic acid, enhances water retention and mechanical stability but inherently slows

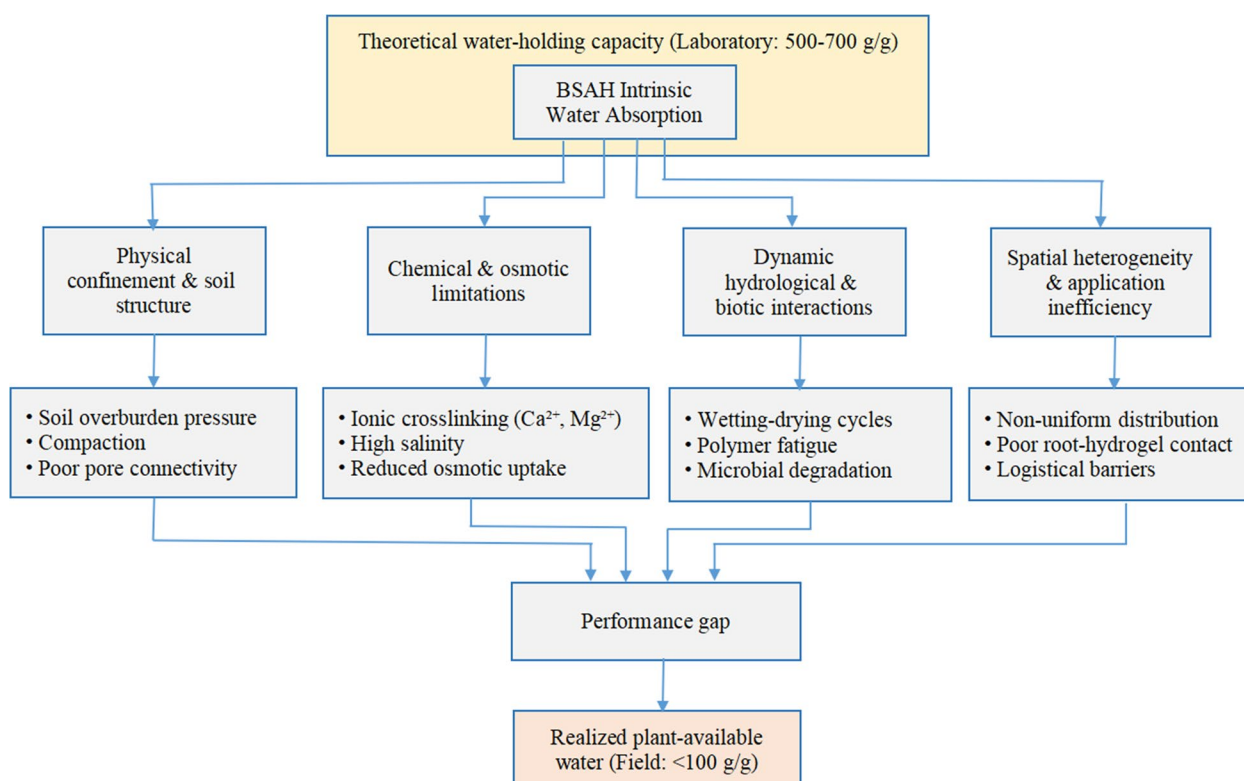


Fig. 6 Field-scale constraints limiting biodegradable superabsorbent hydrogels water absorption efficacy. The model illustrates how soil physical confinement, chemical/osmotic stress, dynamic hydrological

cycles, biotic interactions, and application heterogeneity interact to reduce the realized plant-available water fraction from the theoretical water-holding capacity

Table 1 Comparative swelling performance of biodegradable superabsorbent hydrogels (BSAHs) in distilled water versus controlled soil environments

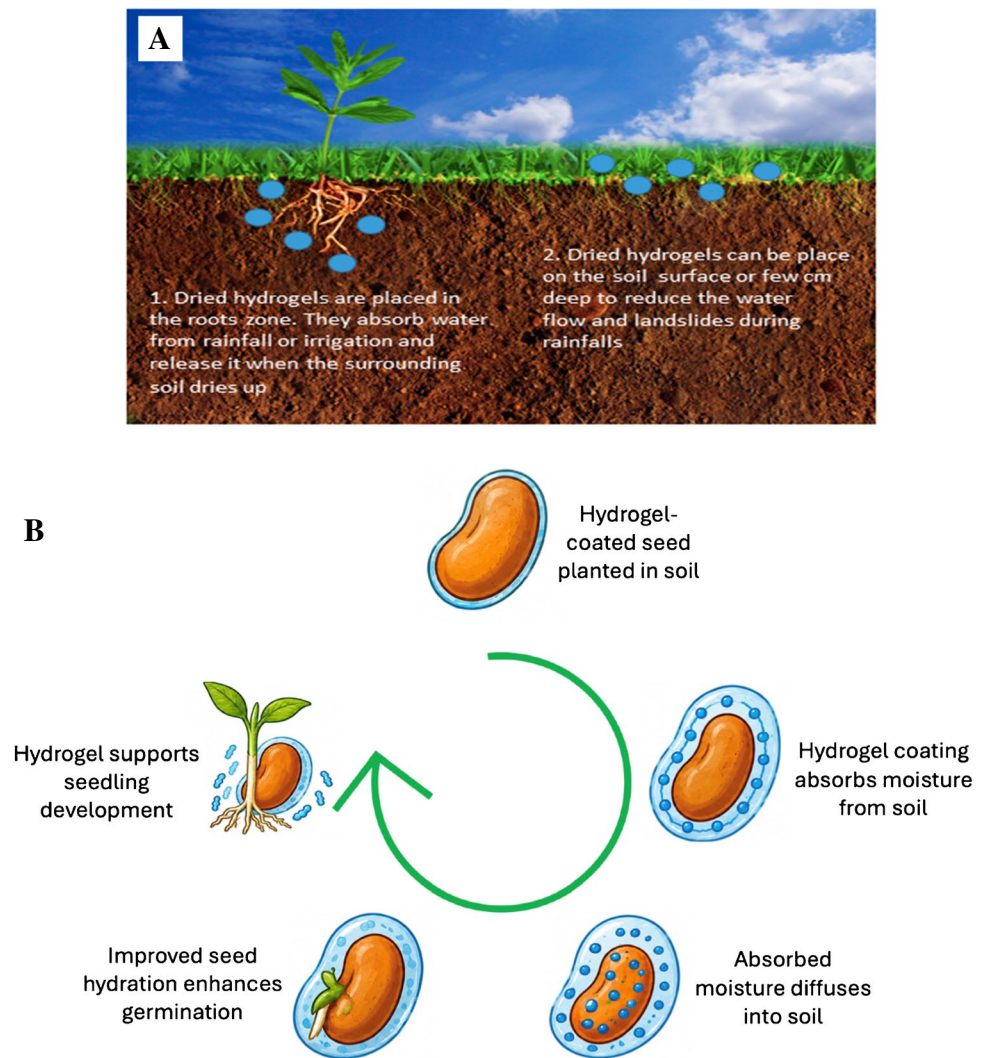
Hydrogel type	Reported swelling (Laboratory, in distilled water)	Reported swelling in soil (Pot/column/greenhouse studies)
Starch-based hydrogel	200–420% in distilled water (Iqbal et al. 2024)	93.11% in loamy-sandy soil (Kathi et al. 2021) 20.2% in soil (Jungsinyatamet al. 2022). 120% in soil (Sarmah and Karak 2020)
Cellulose-based hydrogel	604% in distilled water (Ahmed 2015)	57% in sandy soil (Das et al. 2025) 94.7% (Saha et al. 2020)
Gum-based hydrogel	980% in distilled water (Saleem et al. 2024)	54% in soil (Thombare et al. 2018). 32% in loamy soil (Chen et al. 2024)

biodegradation kinetics. Research indicates that residues from certain acrylamide-based grafts can persist in soil for multiple cropping seasons (Kenawy et al. 2021). This presents not a fundamental contradiction but a deliberate engineering compromise: extending functional longevity in soil comes at the cost of extended environmental residence time. The critical assessment, therefore, shifts from seeking complete biodegradability to evaluating whether the degree and rate of functional degradation are appropriate and environmentally safe for the intended lifespan of the application. Hybrid hydrogels are indeed useful, but their environmental

claims must be calibrated to their specific formulation and validated through long-term field monitoring, rather than inferred from their biopolymer content alone.

In short, this paper suggests that biodegradation of BSAHs is context-dependent, incomplete, and poorly understood. Future research must move beyond simple mass-loss assays and adopt more rigorous approaches, including isotopic carbon tracing, advanced spectroscopic monitoring of breakdown products, and long-term field-scale ecotoxicological assessments. Until such data exist, the assumption that BSAHs represent an environmentally

Fig. 7 Broadcasted hydrogel granules into the soil and incorporated them at planting (A) (Khan et al. 2021); hydrogel seed coating for sustained moisture availability during the germination stage (B) (Created by the authors based on information reported by Kaur et al. (2023)).



benign solution remains more a hypothesis than a proven reality.

3.5 Field-scale implementation techniques

Despite decades of progress in BSAHs design, their translation from laboratory prototypes to scalable agricultural interventions remains limited. Reported application techniques for BSAHs include broadcasting hydrogel granules into the soil (AbdAllah et al. 2021), incorporating them at planting (Kenawy et al. 2021; Khan et al. 2021), and seed coating to improve germination and early seeding vigor (Kaur et al. 2023; Pathak and Ambrose 2020) (Fig. 7a and 7b). Additional approaches also include blending fertilizers to create controlled-release nutrient systems, enhancing nutrient use efficiency while maintaining soil moisture (Dardeer et al. 2024).

Each technique carries trade-offs between cost, uniformity, and agronomic effectiveness. Broadcasting offers

simplicity but suffers from uneven distribution and uncertain root–polymer contact. Seed coating ensures direct root access to hydrogel but is constrained by the small quantity that can be applied without impairing germination (Guilherme et al. 2015; Rakhmetullayeva et al. 2025).

Critically, few multi-season field trials exist, and reported outcomes are often inconsistent or lack agronomic significance. For instance, while small-plot or greenhouse studies often report improved moisture metrics, full-season field trials show inconsistent yield outcomes. A field study on corn under rainfed conditions found that superabsorbent polymers improved early-season moisture but did not translate into significant yield gains under terminal drought (AbdAllah et al. 2021; Nordin et al. 2024). In contrast, a recent field trial with a green hydrogel designed for slow-release urea reported improved wheat yield under deficit irrigation (Etmiani-Esfahani et al. 2025). This highlights that outcomes are highly context-dependent, influenced by hydrogel formulation, crop type, and water management. Similarly, field

evaluations of hydrogel seed coatings have demonstrated variable effectiveness, with benefits often constrained to germination or early growth stages without impacting final harvestable yield under real-world climatic stress. These results underscore that reliable field performance is shaped not only by the hydrogel's intrinsic capacity but also by complex interactions between soil heterogeneity, seasonal climate variability, and application methodology.

In addition, inconsistencies in formulation stability and physical properties, such as swelling behavior, granule size, and degradation rate, pose challenges for uniform mechanical application. Logistical concerns also arise, including transportation and storage of bulky hydrogel materials, especially under humid or wet conditions, which can prematurely activate the product. Moreover, a lack of technical training and awareness among farmers further hampers adoption, particularly in regions where precision agriculture or advanced mechanization is not widely practiced (Toews et al. 2025).

The translational gap from pilot-scale to commercial adoption is stark. While over 85% of published BSAH studies are conducted at laboratory or small-plot scale (Li et al. 2025), fewer than 15% report multi-season field trials with mechanized application. This disparity fundamentally limits extrapolation to commercial farming contexts. Integrating BSAHs into large-scale sowing, irrigation, or fertigation systems remains technically underdeveloped, often requiring customized equipment or retrofitting of existing machinery—an added cost many farmers are unwilling or unable to bear. Without cost-effective, scalable, and standardized deployment methods, the prospect of BSAHs becoming a mainstream agricultural input remains uncertain. Particularly given their negligible current market penetration. Biopolymer-based hydrogels constitute only a minor fraction of the global superabsorbent polymer market, which remains overwhelmingly dominated by conventional, synthetic products (Li et al. 2025; Saha et al. 2020). This study's findings illustrate that while BSAHs hold potential, the field evidence remains fragmented, inconsistent, and technologically constrained. Progress in this area requires not only refining the materials themselves but also designing scalable delivery systems and conducting multi-seasonal experiments.

3.6 Challenges, limitations, and future research directions

The trajectory of BSAH research highlights impressive material innovation but also reveals a persistent gap between laboratory potential and real-world agricultural application. In terms of composition, swelling performance, biodegradation, field deployment, and economics, the literature reveals advances that are scattered and often conflicting, hindering the development of effective, scalable, and environmentally

friendly solutions. A central limitation is the performance–sustainability paradox. As demonstrated by Hasija et al. (2018) and Vishwakarma et al. (2024), grafting synthetic monomers such as acrylamide and acrylic acid onto biopolymers enhances swelling stability and durability under laboratory conditions. However, this improvement comes at the expense of biodegradability, raising the risk of persistent residues and microplastic accumulation in soils (Kenawy et al. 2021; Ni et al. 2023). This trade-off highlights the need to prioritize green crosslinking strategies and biobased additives that strike a balance between performance and environmental safety.

Similarly, as shown in BSAH swelling studies (Guo et al. 2022; Iqbal et al. 2024; Vishwakarma et al. 2024), laboratory-reported water absorption capacities are often between 500 and 700 g/g in distilled water, yet field measurements in real soils rarely surpass 80–100 g/g. This discrepancy is attributed to ionic strength, soil compaction, and salinity variables largely ignored in laboratory protocols. Without standardized agro-climatic testing frameworks, claims of BSAH efficiency remain context-dependent and agronomically uncertain. Biodegradation evidence also suffers from methodological weaknesses. Most studies rely on weight loss or CO₂ evolution as proxies for mineralization (Etminani-Esfahani et al. 2025), which cannot confirm whether hydrogels fully degrade into benign end-products or merely fragment into oligomers. As Kenawy et al. (2021) emphasize, residues from acrylamide-based systems persist in soils, challenging assumptions of environmental safety. Long-term ecological monitoring, utilizing isotopic tracing and metagenomics, is crucial for determining the true environmental fate of BSAHs.

A further critical barrier is the lack of comprehensive techno-economic and life cycle assessments that would allow for a thorough evaluation of BSAH sustainability. Existing techno-economic analyses show that raw material costs for BSAHs are still considerably higher than those for traditional amendments, with cost differences estimated at three to five times. This effectively prices these materials out of most resource-limited farming settings without subsidies (Denagbe et al. 2025). However, these estimates usually omit costs related to field application, scalability, and long-term performance. Agronomic results at the field scale are also inconsistent. AbdAllah et al. (2021) observed early-season moisture improvements in corn but found no significant yield benefit under terminal drought, while Etminani-Esfahani et al. (2025) reported a 12% increase in yield under deficit irrigation with a green hydrogel; neither included cost analysis. Life cycle assessments are also limited. The most comprehensive available LCA data for biobased materials from related sectors, such as packaging and hygiene products, indicate an average 45% reduction in GHG emissions compared to fossil-based equivalents, but these benefits

come with trade-offs, such as a 369% increase in freshwater eutrophication (Zuiderveen et al. 2023). For specific applications like SAPs, Capezza et al. (2023) found that bioSAP production accounted for 78% of total diaper GHG emissions, with only some biobased configurations outperforming fossil-based options. Importantly, no published research has yet combined multi-season field performance data with full cradle-to-grave LCA or techno-economic models specifically for agricultural BSAHs. As a result, claims regarding the economic viability or environmental benefits of BSAHs are not supported by comprehensive, system-level evidence.

From an agronomic standpoint, field application methods such as broadcasting, incorporation, or seed coating have been trialled. While these approaches demonstrate localized benefits, particularly in germination and early growth, their scalability is undermined by labor intensity, lack of mechanized delivery, and inconsistent yield benefits across different agro-ecological zones. Fertigation-based delivery has been conceptually proposed (Li et al. 2024), but remains untested at scale due to technical barriers, such as clogging and viscosity issues.

Finally, techno-economic analyses highlight a critical barrier to adoption. Biopolymer hydrogels remain significantly more expensive than conventional amendments, such as compost or biochar, with production costs limiting their feasibility in smallholder systems (Das et al. 2025; Iqbal et al. 2024; Jungsinayatam et al. 2022; Li et al. 2024). While life cycle assessments suggest that biopolymer-based systems have lower carbon footprints than fully synthetic hydrogels, most LCAs terminate at production and fail to account for downstream agricultural use, persistence, and end-of-life environmental impacts. The mechanistic understanding of field-performance degradation must be integrated with life cycle assessment and economic modeling to create predictive frameworks. Future research should prioritize coupled degradation–efficacy studies that track both material mass loss and water-holding capacity decline under identical field conditions, using advanced techniques like μ CT imaging to visualize pore-clogging and network fatigue (Turioni et al. 2021; Zhou et al. 2025).

These findings underscore the need for integrated, multi-disciplinary frameworks that connect materials science with soil ecology, agronomy, and socio-economic systems. Future research must transcend material optimization and instead adopt whole-system evaluations that measure agronomic effectiveness, ecological safety, and economic viability in tandem. This requires standardized soil–hydrogel testing protocols, multi-season field trials, and full life cycle assessments that capture both environmental and economic trade-offs. This state of the field is critically synthesized in Table 2, which consolidates the unresolved challenges, expert commentary, and priority directions for future research.

Table 2 Critical synthesis of BSAHs across composition, swelling, biodegradation, field implementation, and economic viability: unresolved challenges, critical commentary, and priority future directions

Dimension	Unresolved challenges	Critical commentary	Priority future directions
Composition and Modification	Synthetic grafts reduce the biodegradability of BSAHs, thereby increasing the risk of micro-plastics.	Innovation is often biased toward short-term gains, rather than long-term safety.	Green crosslinking, bio-based additives, field-scale validation.
Swelling and Soil Interaction	Swelling capacity in real soils is thus highly soil specific.	Laboratory metrics systematically overestimate field utility.	Standardized testing under agro-climatic conditions.
Biodegradation and Environmental Fate	Incomplete mineralization, oligomers persist and synthetic additive residues.	The evidence is insufficient for environmental safety claims.	Isotopic tracing, metagenomics, and multi-season monitoring.
Field Implementation	Primarily, pilot-scale, limited mechanization, and inconsistent yields have significant impacts.	Agronomic benefits are short-lived, and scalability is doubtful.	Mechanized delivery systems, multi-site and multi-season validation.
Economic Viability and LCA	High production costs and LCAs are incomplete (ignore field/end-of-life stages).	Sustainability claims are aspirational without cradle-to-grave data.	Integrate techno-economic and ecological modeling.

4 Conclusion

This review critically synthesizes evidence revealing that BSAHs are entrenched in a fundamental and unresolved performance–sustainability paradox. The prevailing research trajectory demonstrates that strategies to enhance agronomic durability and water-retention capacity, typically through synthetic grafting or robust crosslinking, may diminish biodegradability and raise the risk of persistent soil residues. Conversely, formulations designed for benign environmental fate frequently lack the mechanical resilience and hydrological persistence required for reliable crop benefits under field conditions.

Therefore, the field requires a decisive pivot from incremental material science toward integrated, systems-level validation. Merely optimizing swelling capacity in the lab is insufficient. Meaningful progress hinges on two interdependent advances: (1) the non-negotiable application of rigorous tools such as isotopic tracing and metagenomic profiling, to move beyond proxy measures and definitively verify environmental fate and safety; and (2) the parallel development of scalable, mechanized deployment systems that acknowledge the economic and logistical realities of modern agriculture. Ultimately, the long-term viability of BSAHs is not guaranteed by their biobased origins alone. Their credibility as a climate-resilient agricultural input depends critically on embedding these innovations within comprehensive, cradle-to-grave life cycle assessments that transparently weigh agronomic gains against long-term ecological footprints. Without such interdisciplinary and system-oriented rigor, BSAHs risk remaining promising but niche experimental materials, failing to deliver on their potential as scalable and truly sustainable tools for future agriculture.

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